

Agentic AI system for Job Applications and Hiring support for managers

Problem statement and motivation:

In today's extreme job market, where hiring managers and Human Resource departments constantly receive hundreds and thousands of job applications, it is often difficult for managers to manually review each job application due to time constraints and the overflow of applications containing different types of data, such as structured and unstructured data. Smaller companies, like startups in particular who have a lower budget to run their business, do not have the funds to dedicate one employee to the hiring process. This can cause major issues for companies when hiring, such as overlooking qualified candidates, which can lead to hiring the wrong candidate. While many companies collect applicant data through tools such as forms, the actual process of manually reviewing each candidate application and extracting key qualities remains widely manual.

Target users:

- Hiring managers
- HR Professionals
- Small Businesses/startups

Why an agentic AI solution:

- An Agentic AI solution is appropriate for this scenario because, unlike a rule-based system, an agentic system can handle all applicant responses by evaluating unstructured data, deciding which information is important and relevant, and speeding up the process of collecting all valuable information from all applicants, which can heavily influence the quality and speed at which hiring managers and companies recruit candidates. This solution can automatically manage the end-to-end process while also allowing human oversight where key

decisions need to be made, in this case, deciding who to recruit, making the whole process easier and more focused.

Use Case and Scenarios:

Scenario 1: A hiring manager at a small business receives dozens of job applications and wants to carefully select the right candidate, but does not have much time:

- The manager provides the agent with baseline requirements to pass the first level of screening (eg, “the candidate has to be from Georgia”)
- The agent reviews all applications and discards all applications that do not meet the specified requirements
- The agent then extracts important information from applications and structures this information into a Google Sheets document.
- Any missing information is flagged for review.
- Tasks performed by agent: filter and clean form responses, assign unstructured data into structured fields, and validate data for missing values and completeness.

Scenario 2: A HR manager wants a quick review of the top candidates for a job position that has recently opened up to make their job easier.

- The agent reviews all top candidate applications and compares them to the job requirements
- The agent creates a concise summary of the candidate's key strengths, weaknesses, and skills.
- Candidates are categorized and tagged as strong, potential, or weak candidates on the structured data sheet.
- Tasks performed by the agent: Analyzes qualified applicant submissions, compares applicant profiles to job requirements, and generates candidate summaries for the HR manager.

Initial system design:

Preliminary list of agents and their roles:

- Form analysis agent: Analyses job applications
- Data structuring agent: Organizes extracted information into a structured Google Sheets document
- Evaluation agent: evaluates if a candidate is a good fit for the job based on job criteria
- Summarizing agent: summarizes important information regarding the candidate profile for the user.
- Notification agent: notifies the user with alerts and reports as needed.

Tools, API's, or data sources:

- Google Forms API
- Google Sheets API
- Email API (Gmail)
- Large Language Model (such as ChatGPT) for summarization and analysis
- Local or cloud-based storage for submission logs and outputs

Feasibility and risks:

Potential challenges:

- Hallucinations: The LLM may generate incorrect applicant qualifications when inputs are unclear.
- Inconsistent data in submissions: The applicant can provide incomplete or insufficient data in the applications.
- Tool failures: There could be connectivity issues with tools such as Google Sheets and forms.

Constraints:

- Privacy Constraints: Privacy and confidentiality concerns with candidate data
- API rate limits for Google Forms and Google Sheets API
- Time Line for project execution:

- Week 4:
 - Finalize problem definition and scope
 - Define agent roles and responsibilities
 - Design system architecture
 - Identify tools and API's
 - Draft initial prompts for core agents
- Week 5-6:
 - Build a simple interface to interact with the agentic system
- Week 7-8:
 - Transition from UI + Basic logic to an actual agentic system
- Week 9-10:
 - Prepare for formal system evaluation
- Week 11:
 - Review and Final submission